

**EXPERIMENT NO 8****BRAKE TEST ON POLE CHANGING INDUCTION MOTOR****AIM**

- To study the different modes of operation of a 3 phase pole changing induction motor
- Conduct the brake test on 3 phase squirrel cage induction motor and plot the following performance characteristics.
- Electrical characteristics - Speed, line current, torque, power factor, efficiency & % slip Vs output power
- Mechanical characteristics - Speed Vs Torque

APPARATUS

S.No.	Name of the apparatus	Type	Range	Quantity
1.	Voltmeter			
2.				
3.				
4.	Ammeter			
5.				
6.				
7.	Wattmeter			
8.				
9.				
10.	Rheostat			

PRINCIPLE

If an induction motor is to run at different speeds, one way is to have different windings for the motor so that it will have different synchronous speeds and the running speeds. Another method is to use one winding but with suitable connections for a change-over to double the number of poles.

In pole changing induction motors, the stator winding of each phase is divided into two equal groups of coils. These coil groups are connected in series and parallel with the current direction being reversed only in one group, to create two different numbers of poles (even) in the ratio 2:1 respectively. When the connection is changed from series to parallel or vice versa, the current in one group of coils is also reversed at the same time. This technique, termed the consequent pole method, is applied to all three windings (phases). This type of induction motor has always the squirrel cage rotor, which can adapt to any number of stator poles.